

M1 — Replication and Interrogation

Model submission — demonstration paper (Kueh et al., BMJ Open 2026;16:e113719).

Student-facing resolved version. Numerical values in this document were generated from the accompanying course results and match the rendered PDF.

*This is the **Instructor exemplar**: technically correct as far as P1–P5 establish. It is not a planted-error “Review case”.*

Provenance: every result attributed to us is produced or assembled by `M1_code.qmd`. Most are read from structured P1–P5 result objects; M1 additionally runs the replication analysis and the design-aware interaction explicitly identified in the notebook. Figures attributed to the paper are quoted from the article and checked against its PDF.

Slide 1 — The question, and the two contrasts

- **Scientific (locked) estimand — IV vs III.** Among adults meeting FLI-based MASLD criteria, do **non-obese adults with high central adiposity** have higher all-cause mortality than **non-obese adults with low central adiposity**?
- **Replication estimand — IV vs I.** The paper references its contrasts to Group I (obese, low central adiposity). We reproduce that, because reproducing it is what *Replicate* means.
- **Two contrasts from one fitted model.** Re-levelling is reparameterization, not reanalysis. Nothing here says the paper's result is unstable.
- The locked contrast is what we interrogate: it holds body-mass category fixed, so central adiposity carries the exposure.

Slide 2 — The paper reproduces

On the **full analytic file** — the row set the paper itself analyses.

	Published	Ours
Analytic N	~6,300	6,371
Groups I / II / III / IV	386 / 4,541 / 769 / 604	391 / 4,585 / 787 / 608
Deaths	585	586
Group IV vs I, unadjusted	15.13	15.14
Group IV vs I, Model 2	2.893	2.913

- Rebuilt from raw NHANES 2007–2018 plus NCHS linked mortality; no derived extracts.
- **Model 2 is the paper’s headline adjusted model**, fit on the rows complete on its covariates. Every diagnostic that follows uses **Model 1**, so the model label travels with the number everywhere.
- **Four discrepancies logged, none tuned away** — the “BMI <25” header (the Methods text uses 30), PIR percentages that disagree with the counts printed beside them, an alcohol exclusion matching no reported N, and cardiovascular death counted as heart-only (158; adding cerebrovascular gives 185).
- **One thing does not reproduce:** the WHtR spline panel. The knot specification is not reported, so the difference cannot be resolved.

Slide 3 — The analysis population, typed

Step	N	Type
NHANES 2007–2018 records	59,842	—
Age ≥ 18	36,580	ELIGIBILITY
In fasting subsample	14,962	DESIGN
FLI computable	14,170	MISSINGNESS
Steatosis, FLI ≥ 60	6,052	TARGET POPULATION
MASLD criterion	6,050	TARGET POPULATION
Mortality-eligible and linked	6,048	OUTCOME ASCERTAINMENT

- After eligibility, the two largest reductions are **design** and **disease definition**. Neither is missing data.
- **Genuine missingness: 792 item-nonresponse losses. Separately, 2 outcome-ascertainment losses.** Different problems — not added together, and not 30,532, which is what reading every post-age-eligibility reduction as attrition would give.
- **Two row sets, kept apart.** The replication uses the full analytic file (6,371); every diagnostic from here uses the **locked domain, 6,048** — the records with a valid fasting weight.

Slide 4 — The central diagnostic: adjustment versus design

All four cells on the **same 5,939 Model-1 complete-case rows**; *weighted* means WTSAF2YR/6 with strata and PSU.

	IV vs I (<i>replication</i>)	IV vs III (<i>locked</i>)
unweighted, crude	16.05 (5.91–43.58)	2.34 (1.72–3.18)
unweighted, Model 1	3.81 (1.39–10.43)	1.05 (0.77–1.45)
weighted, crude	11.93 (2.60–54.84)	3.07 (2.11–4.48)
weighted, Model 1	2.98 (0.69–12.82)	1.29 (0.89–1.86)

Read **down** a column for adjustment; **across** a row for design.

- **On the locked estimand, adjustment dominates.** Design moves it in the *opposite* direction and by less, and does not materially alter the estimate or the uncertainty statement.
- **Design does materially alter the replication estimate** — 3.81 (1.39–10.43) becomes 2.98 (0.69–12.82) — a lower estimate and a substantially wider interval.
- **The paper’s own Table 2 points the same way.** Its published Model 1 gives Group III 3.175 and Group IV 3.155 against Group I — an implied IV-vs-III ratio of 0.99.

Slide 5 — Interaction: imprecise, and sensitive to design

Targeted **1-df** test of the locked contrast (Groups III and IV only; n = 1,296 — 937 men, 359 women). Both rows are Model 1 on the **same analytic contrast and the same rows**; they differ only in whether the survey design is used.

	ratio of HRs (women / men)	p
unweighted, Model 1	1.00 (0.42–2.35)	0.993
design-aware, Model 1	2.54 (0.92–7.03)	0.073

- **The interaction estimate is sensitive to survey design.** The weighted analysis makes substantial heterogeneity **more plausible** than the unweighted analysis suggested.
- Neither establishes effect modification. **The estimates are too imprecise to resolve the question**, which is not evidence that sex is irrelevant.
- We do not read 0.073 as a finding. The movement in the estimate and interval is the result; the threshold is not.

Slide 6 — Residual missingness: little practical impact here

1 of 11 Model-1 covariates carries any missingness: cancer status, 109 observations (1.80% unweighted, 1.18% weighted). Complete case uses the 5,939 rows complete on Model 1; the other three keep all 6,048.

Locked estimand (IV vs III), weighted Model 1	HR (95% CI)
complete case	1.29 (0.89–1.86)
all assigned no cancer	1.28 (0.89–1.85)
all assigned cancer	1.28 (0.89–1.85)
MI (m = 10)	1.28 (0.89–1.85)

- The four approaches are **materially indistinguishable**, though not identical: complete case gives 1.29 (0.89–1.86) and MI 1.28 (0.89–1.85), a difference of a hundredth, far smaller than either interval.
- The two extreme assignments are transparent global scenarios, **not bounds**: a uniform assignment need not extremise the coefficient.
- Computed on the locked contrast, not inferred from the replication contrast.

Slide 7 — Threats, on two axes

Candidate	What we observed	Consequence
Estimand–adjustment-set alignment	Target quantity never stated; on Slide 4, unweighted, adjustment moves the locked estimand from 2.34 (1.72–3.18) to 1.05 (0.77–1.45)	High
Survey design	Moderate on the locked estimand, large on the replication one	Moderate
FLI circularity	Untested by our pipeline — but not by the authors	Uncertain
Residual missingness	Four approaches materially indistinguishable (Slide 6)	Low

On circularity, be accurate about what already exists. The paper reports a TyG-based re-ascertainment and finds Group III 3.1 (1.3–7.7) and Group IV 3.1 (1.2–7.6) against Group I — so the authors’ own circularity check leaves the locked IV-vs-III contrast at roughly 1.0. **This is a point-estimate observation only; the IV-vs-III uncertainty cannot be recovered from the separately reported confidence intervals.**

Slide 8 — The dominant threat, and what would change our mind

Dominant threat: estimand–adjustment-set alignment. The target quantity Model 1 is meant to represent is not made explicit — causal total effect, causal direct effect, or conditional prognostic association. Without that target, there is no unique causal interpretation of the attenuation, and it is unclear whether the adjustment set supports the paper’s stronger *independent of total body weight* interpretation.

- **Read crudely, Group IV has the poorest observed survival** (19.5% died, against 9.6% in Group III), and the paper reports exactly that. **That descriptive claim does follow.**
- **Read under Model 1, the published Group III and IV point estimates are nearly identical, and our directly estimated IV-vs-III contrast is compatible with no difference** (1.05 (0.77–1.45) unweighted, 1.29 (0.89–1.86) design-aware, both on the Slide 4 rows). The Model 1 results therefore **do not provide clear evidence** that Group IV has higher mortality than Group III. The design-aware interval remains compatible with a meaningful increase as well as with none.

What would change our mind:

- **A stated target quantity**, and — if it is causal — a prespecified causal model supporting the adjustment set. If a direct effect is the target, that requires an explicit mediation estimand.
- **A design-aware re-analysis of the TyG sensitivity on the locked contrast**, with its own interval.
- **A larger or pooled sample** resolving the interaction, since the design-aware Model 1 estimate 2.54 (0.92–7.03) is too imprecise to act on.